

Curriculum Vitae

Personal Information

- **Name:** Dr Ravi Shankar
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Current Positions

- **Position:** Research Fellow
- **Department:** Clinical Research & Innovation Office
- **Institution:** Tan Tock Seng Hospital, National Healthcare Group, Singapore
- **Duration:** October 2025 – Current
- **Position:** Visiting Research Clinician Scientist
- **Department:** Rehabilitation Research Institute of Singapore (RRIS)
- **Institution:** Nanyang Technological University, Singapore
- **Duration:** October 2025 – Current

Past Experience

- **Position:** Research Fellow
- **Department:** Medical Affairs – Research Innovation & Enterprise
- **Institution:** Alexandra Hospital, National University Health System (NUHS), Singapore
- **Duration:** May 2022 – September 2025

Academic Qualifications

- **PhD:** Project Management, National University of Singapore, Aug 2017 - Dec 2021
- **Bachelor's and Master's Degree:** Bachelor and Masters of Technology, Indian Institute of Technology Madras, Chennai, India, Aug 2010 - May 2015

Research Interests and Skills

- **Research Interests:** Health Services Research; Health Policy; Health Technology Assessment; Digital Health Interventions; Implementation Science; Health Economics
- **Skills:** Randomized Controlled Trial Design; Grant Writing; Experience with innovative technologies (AI, VR) in healthcare; Economic Evaluation; Advanced proficiency in Python, STATA, and TREEAGE Pro for data analysis

Total Publications = 88; Average Journal Impact Factor = 4.6; H-Index = 11, i10-index = 15

Google Scholar Link: <https://scholar.google.com/citations?hl=en&user=HhnbeWsAAAAJ>

Academic Website: <https://ravishankarsg.github.io/>

1. **Shankar, R., Ong, R. C. L., Yee, E., Duraisamy, R., Ho, B. C. Y., Sim, R., Kuah, C. W. K., & Ong, P. L. (2026).** Feasibility, acceptability, and functional outcomes of a home-based exergaming telerehabilitation program for adults with functional disabilities: A multi-center single-arm pilot study. *Journal of NeuroEngineering and Rehabilitation*. <https://doi.org/10.1186/s12984-026-02055-x> (Q1, IF: 6.0)
2. **Shankar, R., Lim, A., & Xu, Q. (2026).** Nurses' Lived Experiences of Generative Artificial Intelligence-Enabled Shift Handover Innovation: A Descriptive Phenomenological Study. *Journal of Advanced Nursing*. <https://doi.org/10.1111/jan.70629> (Q1, IF: 3.8)

3. Chandran, G., Kamsani, N. S., Tang, N., Lai, H. S., Ong, T., Cheng, S.-A., Lim, J. H., **Shankar, R., & Chew, E. (2026)**. Case Report: Efficacy of transcutaneous spinal cord stimulation combined with overground robotic-assisted gait training in chronic incomplete spinal cord injury. *Frontiers in Human Neuroscience*. <https://doi.org/10.3389/fnhum.2026.1795068> (Q1, IF: 3.6)
4. **Shankar, R., Tan, C., Devi, F., Ang, E., & Er, J. (2026)**. Nurses' Lived Experiences Managing Peripherally Inserted Central Catheters in Low-Volume Settings: A Phenomenological Study of Professional Liminality. *The Journal of Nursing Research*. (Q1, IF: 2.7)
5. **Shankar, R., Tong, P., Yee, E., & Ong, P. L. (2026)**. Feasibility and usability of a wearable sensor-based telerehabilitation program for home-based exercise in patients with lower limb amputation: A pilot study. *Frontiers in Rehabilitation Sciences*. <https://doi.org/10.3389/fresc.2026.1828171> (Q1, IF: 2.5)
6. **Shankar, R. (2026)**. Sovereign Large Language Models for Structured Data Extraction from Pathology Reports: A Perspective for the Clinical Laboratory. *Laboratories*, 3(3), 9. <https://doi.org/10.3390/laboratories3030009>
7. **Shankar, R., Goh, A., & Xu, Q. (2026)**. Natural language processing-driven insights from social media: Topic modeling and sentiment analysis of healthcare sustainability discourse. *International Journal of Environmental Medicine*, 1(1), 4. <https://doi.org/10.3390/ijem1010004>
8. **Shankar, R., Lim, A., & Xu, Q. (2026)**. Understanding Public Discourse on Alzheimer's Disease and Dementia: A Sentiment Analysis and Topic Modeling Study of Social Media Data. *Journal of Dementia and Alzheimer's Disease*, 3(3), 34. <https://doi.org/10.3390/jdad3030034>
9. **Shankar, R., Chua, K., Krishnan, R. R., Tham, S.-L., Tan, L., Lai, A., Lau, M., Koh, Z. K., Thia, E., Shen, N., & Ong, P. L. (2026)**. Digitization of Rehabilitation Evaluation and Activity Monitoring Using Thermography (DREAMT): Protocol for a mixed-methods randomized multiple baseline study in inpatient rehabilitation. *Frontiers in Digital Health*. <https://doi.org/10.3389/fdgth.2026.1884320> (Q1, IF: 5.2)
10. **Shankar, R., Sundar, V., Goh, Z., Sin, P. Y., & Tan, E. Y. (2026)**. Performance of large language models for extracting clinical data from breast cancer pathology reports: A systematic review. *npj Digital Medicine*. <https://doi.org/10.1038/s41746-026-02789-x> (Q1, IF: 18.0)
11. **Shankar, R., Devi, F., Kee, K., Low, E., & Ali, J. (2026)**. Experiences of Stigma, Bias, and Communication Challenges Among Pregnant Healthcare Workers: A Systematic Review of Qualitative Evidence. *International Journal of Nursing Studies*. <https://doi.org/10.1016/j.ijnurstu.2026.105629> (Q1, IF: 8.7)
12. **Shankar, R., Wang, L., Ho, S. H., Liew, M. F., Kumar, S. P., Wong, T. C., & Wong, S.** Cost-Effectiveness of Virtual Emergency Care Models: A Systematic Review. *JMIR mHealth and uHealth*. (Q1, IF: 6.3)
13. **Shankar, R., Lim, A., & Xu, Q. (2026)**. Performance of Large Language Models in Data Extraction for Evidence Synthesis: A Systematic Review. *Journal of Biomedical Informatics*. (Q1, IF: 5.9)
14. **Shankar, R., Wang, L., Hoe, H. S., Lee, I. Y., Liew, M. F., Gollamudi, S. P. K., Wong, T. C., & Wong, S. (2026)**. The role of artificial intelligence in virtual emergency care: a systematic review. *International Journal of Medical Informatics*. <https://doi.org/10.1016/j.ijmedinf.2026.106411> (Q1, IF: 5.0)
15. **Shankar, R., Goh, Z., & Xu, Q. (2026)**. Techniques, performance, and feasibility of natural language processing for abstract screening in evidence synthesis: A systematic review. *Campbell Systematic Reviews*. <https://doi.org/10.1177/18911803261469813> (Q1, IF: 3.8)

16. Yap, E. Y. A., Chia, D. B., Lee, J. J.-M., Choo, M. H., Cheng, H. K., & **Shankar, R.** (2026). The Effectiveness of Psychoeducational Interventions for Family Caregivers of People with Dementia in Asia: A Systematic Review. *Journal of Alzheimer's Disease*. <https://doi.org/10.1177/13872877261461174> (Q2, IF: 3.4)
17. **Shankar, R.**, Devi, F., Joshua, J. W., & Ng, T. M. (2026). Barriers and enablers to pharmacist involvement in social prescribing: a systematic review of qualitative studies. *Research in Social and Administrative Pharmacy*. <https://doi.org/10.1016/j.sapharm.2026.01.005> (Q1, IF: 3.4)
18. **Shankar, R.**, Lim, A., & Xu, Q. (2026). Barriers and Facilitators to the Use of Large Language Model-Based Conversational Agents in Mental Healthcare: A Systematic Review. *Healthcare*. <https://doi.org/10.3390/healthcare14101267> (Q1, IF: 3.4)
19. **Shankar, R.** (2026). Large language models in post-acute and long-term care: Bridging promise and practice. *Journal of the American Medical Directors Association*. <https://doi.org/10.1016/j.jamda.2026.106379> (Q1, IF: 3.4)
20. **Shankar, R.**, Lo, Y. T., Fong, C. L., Keong, N., & Chua, K. S. G. (2026). Motor imagery brain-computer interface (MI-BCI)-assisted upper limb neurorehabilitation for acute stroke during inpatient rehabilitation: A prospective feasibility study with economic evaluation protocol. *Journal of Clinical Medicine*. <https://doi.org/10.3390/jcm15145692> (Q1, IF: 3.3)
21. **Shankar, R.**, Ong, P. L., & Chua, K. S. G. (2026). Cost-effectiveness of home-based fall detection and monitoring technologies for community-dwelling older adults: A systematic review protocol. *BMJ Open*. <https://doi.org/10.1136/bmjopen-2026-117122> (Q1, IF: 2.5)
22. **Shankar, R.**, Lim, A., & Xu, Q. (2026). Effectiveness of Digital Tools in Supporting Young Caregivers: A Systematic Review. *Early Intervention in Psychiatry*. <https://doi.org/10.1111/eip.70215> (Q3, IF: 2.3)
23. **Shankar, R.**, Goh, Z., Xu, Q., Chew, E., & Chua, K. S. G. (2026). Perceptions, attitudes, and lived experiences of therapists with lower limb robotic exoskeletons in stroke rehabilitation: A systematic review of qualitative studies. *Disability and Rehabilitation*. <https://doi.org/10.1080/09638288.2026.2637191> (Q2, IF: 2.2)
24. **Shankar, R.**, Tan, A., & Xu, Q. (2026). Lived experiences of patients using left ventricular assist devices (LVADs): a systematic review of qualitative studies. *Discover Public Health*. <https://doi.org/10.1186/s12982-026-01907-0> (Q4, IF: 1.3)
25. **Shankar, R.**, Sundar, V., & Chua, K. S. G. (2026). Barriers and Facilitators to the Adoption of Speech-Based Cognitive Screening Tools: A Protocol for Systematic Review of Qualitative Evidence. *Campbell Systematic Reviews*. <https://doi.org/10.1177/18911803261460855> (Q1, IF: 3.8)
26. **Shankar, R.**, Kumar, F. D. S., Bunde, A., & Mukhopadhyay, A. (2026). Effectiveness and user experience of ambient artificial intelligence-driven scribe technology among clinicians: A protocol for a systematic review. *Systematic Reviews*. <https://doi.org/10.1186/s13643-026-03220-y> (Q1, IF: 5.7)
27. **Shankar, R.**, Bunde, A., Low, J., & Hong, W. Z. (2026). Barriers and facilitators to dietary adherence in adults with chronic kidney disease: a protocol for systematic review. *Systematic Reviews*. <https://doi.org/10.1186/s13643-026-03080-6> (Q1, IF: 5.7)
28. **Shankar, R.**, Lim, A., & Xu, Q. (2026). Emergency Nurses' Experiences With Artificial Intelligence-Driven Triage Clinical Decision Support Systems: A Protocol for a Systematic Review of Qualitative Studies. *Journal of Emergency Nursing*. <https://doi.org/10.1016/j.jen.2026.05.010> (Q1, IF: 3.2)
29. **Shankar, R.**, Shafawati, N., & Chandran, G. (2026). Effectiveness of overground robotic exoskeletons in early inpatient rehabilitation for incomplete spinal cord injury: A

systematic review protocol. *Frontiers in Neuroscience*.

<https://doi.org/10.3389/fnins.2026.1781656> (Q2, IF: 4.0)

30. **Shankar, R., Tang, N., & Chandran, G. (2026).** Combined Repetitive Transcranial Magnetic Stimulation and Transcutaneous Spinal Cord Stimulation for Upper Limb Recovery in Chronic Incomplete Cervical Spinal Cord Injury: Protocol for a Pilot Randomized Controlled Trial. *Frontiers in Neuroscience*.
<https://doi.org/10.3389/fnins.2026.1833493> (Q2, IF: 4.0)
31. **Shankar, R., & Chandran, G. (2026).** Lower Limb Rehabilitation in Chronic Incomplete SCI: A Randomized Controlled Trial Protocol Comparing Combined TMS-tSCS Neuromodulation Versus tSCS Alone. *Frontiers in Neuroscience*.
<https://doi.org/10.3389/fnins.2026.1773917> (Q2, IF: 4.0)
32. **Shankar, R., Sundar, V., & Tay, M. R. J. (2026).** Barriers and facilitators to uptake of non-surgical interventions for knee osteoarthritis: A protocol for a systematic review of qualitative studies. *Frontiers in Medicine*. <https://doi.org/10.3389/fmed.2026.1791068> (Q1, IF: 3.6)
33. **Shankar, R., Devi, F., & Xu, Q. P. (2026).** Effectiveness of Digital Tools in Promoting Advance Care Planning: A Systematic Review Protocol. *Frontiers in Medicine*.
<https://doi.org/10.3389/fmed.2025.1698865> (Q1, IF: 3.6)
34. **Shankar, R. (2026).** Comment on ‘Antenatal lifestyle interventions to reduce gestational weight gain: where should we start and how much will it cost?’. *Public Health*. <https://doi.org/10.1016/j.puhe.2026.106422> (Q1, IF: 3.5)
35. **Shankar, R., Choo, S. X., Chen, Z., Kuah, C. W. K., Plunkett, T. K., Ng, C. Y., Lin, S., Goh, K. H., Yee, E., Ge, X., Zhang, D., Chong, W. B., Low, J. A. M., Lau, M. S. E., Lim, X. Y., Naing, S. Y., Wong, L. T., Noronha, B., Aguirre-Ollinger, G., Hussain, A., Ong, P. L., & Chua, K. S. G.** Telerehabilitation Robotics for Upper Limb Rehabilitation after Stroke (TRUST): A Multi-Site Pragmatic Trial Protocol. (2026). *Frontiers in Neurology*. <https://doi.org/10.3389/fneur.2026.1775829> (Q2, IF: 3.3)
36. **Shankar, R., Hue, Y. L., & Chua, K. S. G. (2026).** Neurosurgical Intervention versus Conservative Management in Older Adults with Traumatic Brain Injury: A Protocol for a Systematic Review and Meta-analysis. *Frontiers in Neurology*.
<https://doi.org/10.3389/fneur.2026.1775791> (Q2, IF: 3.3)
37. **Shankar, R., Lim, A., & Xu, Q. (2026).** Cardiovascular and cerebrovascular safety of calcitonin gene-related peptide (CGRP)-targeted monoclonal antibodies and gepants for the preventive treatment of migraine in patients with pre-existing vascular risk factors: A systematic review and meta-analysis protocol. *SN Comprehensive Clinical Medicine*.
<https://doi.org/10.1007/s42399-026-02446-0>
38. **Shankar, R., Lim, A., & Xu, Q. (2026).** A qualitative systematic review protocol of patient experiences of using large language models for health information seeking and healthcare decision making. *Discover Public Health*. <https://doi.org/10.1186/s12982-026-02067-x> (Q4, IF: 1.3)
39. **Shankar, R., Lim, A., & Xu, Q. (2026).** A systematic review protocol of qualitative studies on lived experiences of stigmatization in men with eating disorders. *Discover Public Health*. <https://doi.org/10.1186/s12982-026-02230-4> (Q4, IF: 1.3)
40. **Shankar, R. (2026).** The Jevons trap: when artificial intelligence in healthcare creates endless work. *The American Journal of Medicine*.
<https://doi.org/10.1016/j.amjmed.2026.05.010> (Q1, IF: 6.0)
41. **Shankar, R. (2026).** Beyond the package insert: Operationalising lower clozapine doses for Asian patients in routine clinical practice. *Asian Journal of Psychiatry*.
<https://doi.org/10.1016/j.ajp.2026.105036> (Q1, IF: 4.9)
42. **Shankar, R., & Xu, Q. (2026).** Advancing Artificial Intelligence in Renal Nutrition: The Need for Culturally Adaptive, Equity-Focused Frameworks Beyond Retrieval-

- Augmented Generation. *Journal of Renal Nutrition*.
<https://doi.org/10.1053/j.jrn.2026.04.016> (Q1, IF: 4.0)
43. **Shankar, R. (2026)**. Navigating Professional Accountability in AI-Assisted Nursing Practice: Ethical and Legal Imperatives for the Digital Age. *SAGE Open Nursing*.
<https://doi.org/10.1177/23779608261457722> (Q1, IF: 2.9)
 44. **Shankar, R., & Xu, Q. (2026)**. Beyond accuracy: advancing LLM evaluation in clinical neurology toward real-world decision support. *Neurological Sciences*.
<https://doi.org/10.1007/s10072-026-09125-x> (Q3, IF: 2.7)
 45. **Shankar, R., Bunde, A., & Mukhopadhyay, A. (2026)**. Explainable Artificial Intelligence in Speech-Based Cognitive Decline Detection: A Systematic Review. In J. M. Juarez et al. (Eds.), *Explainable Artificial Intelligence and Process Mining Applications for Healthcare* (Communications in Computer and Information Science, Vol. 2749, pp. 40–55). Springer, Cham. https://doi.org/10.1007/978-3-032-21514-7_5
 46. Xu, P., Xu, J., Clark, J., & **Shankar, R. (2026)**. AI-enabled climate-health nexus framework for decision-support in sustainable urban green infrastructure. *Proceedings of the 43rd International Symposium on Automation and Robotics in Construction (ISARC 2026)*, 896–900. <https://doi.org/10.22260/ISARC2026/0115> (Conference Proceedings)
 47. **Shankar, R., & Mukhopadhyay, A. (2025)**. Social media insights on sepsis management using advanced natural language processing techniques. *Critical Care*, 29(1), 115. DOI: <https://doi.org/10.1186/s13054-025-05344-4> (Q1, IF: 11.4)
 48. **Shankar, R., Qian, X., & Bunde, A. (2025)**. Patient voices in dialysis care: A sentiment analysis and topic modeling study of social media discourse. *JMIR*. DOI: <https://doi.org/10.2196/70128> (Q1, IF: 8.2)
 49. Sumner, J., **Shankar, R.**, Bunde, A., Yap, A., Mohamed Ali, J., Teng, G. G., Phang, K. F., Yip, A. W., & Lim, Y. W. (2025). Insights from high and low clinical users of telemedicine: A mixed-methods study of clinician workflows, sentiments, and user experiences. *International Journal of Medical Informatics*, 184, 106044. DOI: <https://doi.org/10.1016/j.ijmedinf.2025.106044> (Q1, IF: 5.0)
 50. **Shankar, R., & Yip, A. (2025)**. Sentiment analysis and topic modeling of social media data to explore public discourse on irritable bowel syndrome. *Scientific Reports*, 15(1), 10829. DOI: <https://doi.org/10.1038/s41598-025-08599-7> (Q1, IF: 4.9)
 51. **Shankar, R., & Tan, W. (2025)**. Strategic stakeholder management in public–private partnerships. *Engineering, Construction and Architectural Management*. DOI: <https://doi.org/10.1108/ECAM-12-2024-1681> (Q1, IF: 4.5)
 52. **Shankar, R., Tang, N., Shafawati, N., Phan, P., Mukhopadhyay, A., & Chew, E. (2025)**. Cost-effectiveness analysis of robotic exoskeleton versus conventional physiotherapy for stroke rehabilitation in Singapore from a health system perspective. *BMJ Open*. DOI: <https://doi.org/10.1136/bmjopen-2024-095269> (Q2, IF: 2.5)
 53. **Shankar, R., & Yip, A. (2025)**. Transforming patient feedback into actionable insights through natural language processing: A knowledge discovery and action research study. *JMIR Formative Research*. DOI: <https://doi.org/10.2196/69699> (Q2, IF: 2.4)
 54. **Shankar, R., Bunde, A., & Mukhopadhyay, A. (2025)**. Natural language processing of electronic health records for early detection of cognitive decline: a systematic review. *npj Digital Medicine*, 8(1), 133. DOI: <https://doi.org/10.1038/s41746-025-01527-z> (Q1, IF: 18.0)
 55. **Shankar, R., Kannan, J., Tan, Y. H., & Xu, Q. (2025)**. Natural Language Processing Techniques to Detect Delirium in Hospitalized Patients from Clinical Notes: A Systematic Review. *npj Digital Medicine*. <https://doi.org/10.1038/s41746-025-02051-w> (Q1, IF: 18.0)

56. **Shankar, R., Goh, Z., Devi, F., & Xu, Q. (2025).** A systematic review of explainable artificial intelligence methods for speech-based cognitive decline detection. *npj Digital Medicine*, 8, 724. DOI: <https://doi.org/10.1038/s41746-025-02105-z> (Q1, IF: 18.0)
57. **Shankar, R., Bundele, A., & Mukhopadhyay, A. (2025).** A systematic review of natural language processing techniques for early detection of cognitive impairment. *Mayo Clinic Proceedings: Digital Health*, 3(2), 100205. DOI: <https://doi.org/10.1016/j.mcpdig.2025.100205> (Q1, IF: 5.0)
58. **Shankar, R., Wen, K. S. W., Goh, Z., Tan, B., & Chandran, G. (2025).** Effectiveness of transcutaneous spinal cord stimulation for lower limb rehabilitation in spinal cord injury: A systematic review. *Journal of NeuroEngineering and Rehabilitation*. <https://doi.org/10.1186/s12984-025-01775-w> (Q1, IF: 6.0)
59. **Shankar, R., Lee, E. W. X., Luo, N., Khatri, P., Leong, L., Mukhopadhyay, A., Chua, H.-R., & Hong, W.-Z. (2025).** Assessing caregiver burden in kidney failure: A systematic review of measurement properties of instruments. *Kidney Medicine*, 7(3), 100456. DOI: <https://doi.org/10.1016/j.xkme.2025.101054> (Q1, IF: 4.1)
60. **Shankar, R., Foo, T. H., Devi, F., & Xu, Q. (2025).** Patient perspectives on AI-powered cognitive behavioral therapy tools in managing anxiety and stress: A systematic review of qualitative studies. *Journal of Mental Health*. DOI: <https://doi.org/10.1080/09638237.2025.2595612> (Q1, IF: 3.5)
61. **Shankar, R., & Mukhopadhyay, A. (2025).** Effectiveness of green practices in intensive care units to promote environmental sustainability: A systematic review. *Indian Journal of Critical Care Medicine*, 29(S1), S287. DOI: <https://doi.org/10.5005/jaypee-journals-10071-24933.221> (Q3, IF: 2.0)
62. **Shankar, R., Devi, F., & Xu, Q. (2025).** A systematic review of the application of computational grounded theory method in healthcare research. *Biology Methods & Protocols*, bpaf088. DOI: <https://doi.org/10.1093/biomethods/bpaf088> (Q3, IF: 2.4)
63. **Shankar, R., Chew, E., Bundele, A., & Mukhopadhyay, A. (2025).** Protocol for detection and monitoring of post-stroke cognitive impairment through AI-powered speech analysis: A mixed methods pilot study. *Frontiers in Aging Neuroscience*. DOI: <https://doi.org/10.3389/fnagi.2025.1581891> (Q1, IF: 5.2)
64. **Shankar, R., Siva Kumar, F. D., & Mukhopadhyay, A. (2025).** The role of leadership in fostering psychological safety in healthcare: Protocol for a systematic review of qualitative studies. *International Journal of Qualitative Methods*. DOI: <https://doi.org/10.1177/16094069251333895> (Q1, IF: 7.2)
65. **Shankar, R., Devi, F., Er, J., Ang, E., Lam, J. M. P., & Mukhopadhyay, A. (2025).** From ward nurse to home-based practitioner: A protocol for a hermeneutic phenomenological inquiry into evolving professional identity. *International Journal of Qualitative Methods*. DOI: <https://doi.org/10.1177/16094069251337867> (Q1, IF: 7.2)
66. **Shankar, R., Devi, F., Ali, J., et al. (2025).** Experiences of stigma, bias, and communication challenges among pregnant healthcare workers: A protocol for a systematic review of qualitative evidence. *International Journal of Qualitative Methods*. DOI: <https://doi.org/10.1177/16094069251370654> (Q1, IF: 7.2)
67. **Shankar, R., Bundele, A., Yap, A., et al. (2025).** Development and feasibility testing of an AI-powered chatbot for early detection of caregiver burden: Protocol for a mixed methods feasibility study. *Frontiers in Psychiatry*, 16. <https://doi.org/10.3389/fpsy.2025.1553494> (Q2, IF: 3.8)
68. **Shankar, R., Siva Kumar, F. D., Bundele, A., et al. (2025).** Virtual reality for stress management and burnout reduction in nursing: A systematic review protocol. *PLOS ONE*, 20(4), e0319247. DOI: <https://doi.org/10.1371/journal.pone.0319247> (Q2, IF: 2.8)

69. **Shankar, R., Bundele, A., & Mukhopadhyay, A. (2025).** Barriers and enablers for the deployment of large language model-based conversational robots for older adults: A protocol for a systematic review of qualitative studies. PLOS ONE, 20(4), e0321093. DOI: <https://doi.org/10.1371/journal.pone.0321093> (Q2, IF: 2.8)
70. **Shankar, R., Wang, L., Hoe, H. S., Fong, L. M., Gollamudi, S. P. K., & Wong, S. (2025).** Cost-Effectiveness of Virtual Emergency Care Models: A protocol for a Systematic Review. PLOS ONE, 20(4), e0330946. DOI: <https://doi.org/10.1371/journal.pone.0330946> (Q2, IF: 2.8)
71. **Shankar, R., Devi, F., & Mukhopadhyay, A. (2025).** Generalist-led hospital models and their alignment with value-based care: A systematic review protocol. BMJ Open. DOI: <https://doi.org/10.1136/bmjopen-2025-100238> (Q2, IF: 2.5)
72. **Shankar, R., Bundele, A., Low, J., et al. (2025).** Barriers and enablers to pharmacist involvement in social prescribing: A protocol for a systematic review of qualitative studies. BMJ Open, 15(2), e099022. DOI: <https://doi.org/10.1136/bmjopen-2025-099022> (Q2, IF: 2.5)
73. **Shankar, R., Devi, F., Ang, E., & Er, J. (2025).** Barriers and facilitators to nurses' adoption of artificial intelligence-driven solutions in clinical practice: A protocol for a systematic review of qualitative studies. BMJ Open. DOI: <https://doi.org/10.1136/bmjopen-2025-099875> (Q2, IF: 2.5)
74. **Shankar, R., Bundele, A., Yap, A., & Mukhopadhyay, A. (2025).** Experiences and user perceptions of virtual reality-based mindfulness interventions: Protocol for a systematic review and thematic synthesis. BMJ Open. DOI: <https://doi.org/10.1136/bmjopen-2024-094876> (Q2, IF: 2.5)
75. **Shankar, R., Wang, L., Hoe, H. S., Fong, L. M., Gollamudi, S. P. K., & Wong, S. (2025).** The role of artificial intelligence in virtual emergency care: A protocol for a systematic review. BMJ Open. DOI: <https://doi.org/10.1136/bmjopen-2025-103084> (Q2, IF: 2.5)
76. **Shankar, R., Devi, F., & Qian, X. (2025).** Virtual reality interventions to reduce social isolation in older adults: A protocol for systematic review. BMJ Open. DOI: <https://doi.org/10.1136/bmjopen-2025-100830> (Q2, IF: 2.5)
77. **Shankar, R., Bundele, A., & Mukhopadhyay, A. (2025).** Effectiveness of chatbot interventions for reducing caregiver burden: Protocol for a systematic review and meta-analysis. MethodsX, 14, 103272. DOI: <https://doi.org/10.1016/j.mex.2025.103272> (Q2, IF: 2.9)
78. **Shankar, R., Bundele, A., & Mukhopadhyay, A. (2025).** The effectiveness of virtual reality-based mindfulness interventions for managing stress, anxiety, and depression: Protocol for a systematic review and meta-analysis of randomized controlled trials. JMIR Research Protocols. DOI: <https://doi.org/10.2196/68231> (Q3, IF: 1.6)
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Research and Innovation Projects (Total: SGD 672,663)

Funder/ Grantor Name	Grant Program Name	Role	Title	Total Project Value (DC + Salary Support + IRC)
NMRC	Healthy Longevity Catalyst Awards (HLCA) 2024	Principal Investigator	Potential of Artificial Intelligence to Track and Predict Cognitive Impairment in Stroke Rehabilitation	SGD 164,450
NUHS	NUHS Seed Grant Call - NUHS Seed Fund Junior (Mar 2024)	Principal Investigator	Enhancing Psychosocial Well-Being of End-Stage Kidney Disease Caregivers Through Virtual Reality-Based Mindfulness Interventions	SGD 140,820
Synapxe	Synapxe - GenAIus Challenge	Principal Investigator	Development and Evaluation of an AI driven Nursing Shift Handover Assistant	SGD 98,093
National University of Singapore	Academic Health Programme (AHP) Fund 2024	Principal Investigator	An AI-Driven Chatbot System (BOTANIC) for the Early Detection of Caregiver Burden in Caregivers of End-Stage Kidney Disease: A feasibility study	SGD 49,200
NUHS	NUHS Health Services Research (HSR) Seed Grant (Sept 2023 Call)	Co-Investigator	Comparing the level of agreement amongst a self-reported questionnaire, trained health coach evaluation, and sentiment analysis in classifying patients' readiness	SGD 220,100